CIS I (601.455/655) — Assignment 2: Kinematic Calibration of a Cartesian Robot

Design, implementation and results

Zijun Xiang, [teammate]

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1 Problem statement and goals

A Cartesian robot (three orthogonal linear stages) carries a spherical wrist and a tool holder. Its controller knows only the *nominal* kinematics

$$\mathbf{F}_{ee}(\vec{\theta}) = \mathrm{Trans}(\theta_1\vec{d}_1+\theta_2\vec{d}_2+\theta_3\vec{d}_3)\,\mathrm{Trans}(500,0,0)\,\mathbf{R}(\vec{n}_4,\theta_4)\mathbf{R}(\vec{n}_5,\theta_5)\mathbf{R}(\vec{n}_6,\theta_6)\,\mathrm{Trans}(100,0,0),$$

while the real robot has tilted axes ($\vec{d}_k^{\,*} = \mathbf{R}(\vec{\phi}_k)\vec{d}_k$, $\|\vec{\phi}_k\| \le 0.02$ rad for $k = 2\dots6$), systematic and random joint errors ($\theta_k^* = \theta_k + \Delta\theta_k^{sys}(\theta_k) + \Delta\theta_k^{rnd}$), a tool-frame error ($\mathbf{F}_{we}^* = \mathbf{F}_{we}\Delta\mathbf{F}_{ee}$) and an uncertain base pose ($\mathbf{F}_{rob}^* = \Delta\mathbf{F}_{rob}\mathbf{F}_{rob}$). Uncalibrated, the end effector is several millimetres from where the controller thinks it is (Table 8: 7.070 mm rms, up to 16.10 mm and 5.60e-02 rad on the test grid).

An optical tracker observes a calibration marker body whose coordinate system *is* $\mathbf{F}_{ee}$. Our program must design the scene, collect tracker data against a simulated clock, and deliver `ImprovedFwdKins` / `ImprovedInvKins` that are accurate to $\|\vec{\epsilon}\| \le 0.2$ mm and $\|\vec{\alpha}\| \le 10^{-3}$ rad over the whole work volume, in as little simulated time as possible (soft budget 900 s), with the program itself finishing in under 10 minutes of real time.

Result in one line. Method 1 of the handout (one joint at a time from home) with a 31-parameter least-squares model reaches 0.118 mm / 3.48e-04 rad worst case for `ImprovedFwdKins` and 0.115 mm / 4.94e-04 rad for `ImprovedInvKins` on the fixed 729-pose test grid (seed 1), using 389 s of simulated calibration time; all 3 seeds tried meet both targets (Table 8).

2 Scene design (Step 1)

2.1 Where the markers can be

The wrist centre reaches $x \in [500, 1100]$, $y \in [-300, 300]$, $z \in [0, 600]$ mm (base coordinates). The end effector is 100 mm beyond the wrist centre and the calibration markers at most 75 mm beyond that, so **every marker of every reachable pose lies inside a cube of half-size 475 mm centred at (800, 0, 300) mm** — the *marker envelope*. We design visibility for the whole envelope rather than for a grid of poses, so no pose needs to be checked at run time.

2.2 Tracker pose

The viewing volume (handout Figure 2) is a frustum along the camera's $+z$ axis: depths 1000–2200 mm, half-width 500 mm at the near face growing to 1000 mm at the far face. We place the camera **1625 mm above the envelope centre, looking straight down** (`design.tracker_pose()`):

Table 1: Visibility margins of the marker envelope for the chosen tracker pose.

quantity	value
envelope depth range	1150 – 2100 mm (frustum: 1000 – 2200 mm)
lateral margin at the envelope's near face	562.5 − 475 = **87.5 mm**
depth margin to the far face / near face	**100 mm** / 150 mm
worst-case shift of the real base ($0.05\,\mathrm{rad} \times 1566\,\mathrm{mm} + 5\,\mathrm{mm}$)	83 mm

The margin exceeds the worst-case placement error of the robot base, so the body is visible at every pose for every admissible $\Delta\mathbf{F}_{rob}$. The handout sets the tracker jiggle to zero, so this pose is

exact and the tracker frame is our fixed reference. Moving the camera closer would lose lateral margin at the near face; further would lose depth margin (Figure 1).

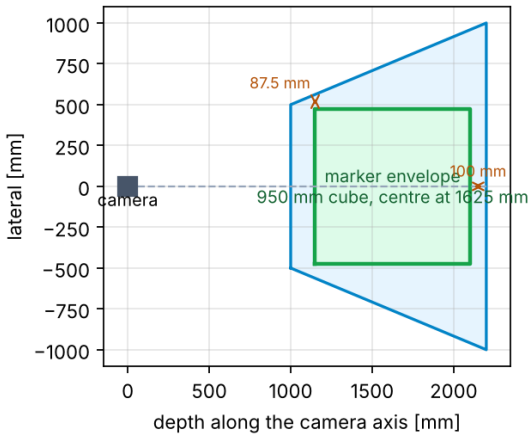


Figure 1: Side view (to scale) of the viewing volume with the marker envelope at 1625 mm.

2.3 Robot pose

$\mathbf{F}_{rob} = \mathbf{I}$: the base is placed at the workspace origin. The choice is arbitrary because the real base pose is one of the calibrated parameters.

2.4 Calibration body

Eight markers on the corners of a cube with circumradius 74.99 mm (the largest allowed is 75 mm; the 0.01 mm keeps the corner coordinates inside the limit after rounding), `design.mbee_layout()`. With isotropic marker noise σ_b the orientation error of one registration about an axis $\vec{u}$ is $\sigma_b / \sqrt{\sum_k |\vec{a}_k \times \vec{u}|^2}$; for this layout the sum is $8 \cdot 75^2 \cdot \frac{2}{3} = 30000$ mm^2 for every $\vec{u}$ (the layout is isotropic), giving 5.8×10^{-4} **rad** and $0.1/\sqrt{8} = 0.035$ mm per reading. A six-marker octahedron of the same radius gives 6.7×10^{-4} rad; the cube is the best use of the allowed size. We define no additional marker bodies: there is no tracker jiggle to compensate, and their layouts are only known to ±2 mm.

3 Error model and error budget

3.1 Which specifications matter

Table 2: The spec sheet, sorted by what it does to the error budget.

spec-sheet item	value	effect if ignored	how we handle it
axis tilts $\vec{\phi}_k$	≤ 0.02 rad	up to $0.02 \times 600 = 12$ mm and 0.02 rad	10 tilt parameters (2 per axis)
joint distortion, wrist	≤ 0.01 rad, slope ≤ 5×10^{-5}	1 mm at the tool, 0.01 rad	homing offset + linear term
joint distortion, slides	≤ 0.2 mm, slope ≤ 5×10^{-4}	0.2 mm	homing offset + quadratic

spec-sheet item	value	effect if ignored	how we handle it
tool frame $\Delta\mathbf{F}_{ee}$	≤ 0.01 rad, 0.5 mm	0.5 mm, 0.01 rad everywhere	6-parameter tool twist
base placement	≤ 0.05 rad, 5 mm	only matters for visibility	6-parameter base twist (5 fitted)
joint random error	σ = 0.005 mm, 10^{-4} rad	≈ 0.02 mm, 2×10^{-4} rad	cannot be calibrated: error floor
tracker noise σ_b	0.1 mm / marker	5.8×10^{-4} rad / reading	average M readings per pose
tracker systematic field	≤ 0.1 mm, gradient ≤ 7.5×10^{-4}	≤ 0.1 mm per point, slowly varying	cannot be calibrated: error floor
tracker jiggle	0	—	—
motion limits	50 mm/s, 0.1 rad/s	sets the time	plan order, sweep length

3.2 Sample spacing from the slope bounds (method 1, step a)

If an error $e(\theta)$ has slope $|e'| \le \omega$ and is sampled every h, linear interpolation between samples is off by at most $\omega h/2$ (the worst case is a "tent" between two samples). Giving interpolation one tenth of the position budget ($\varepsilon = 0.02$ mm) requires $h \le 2\varepsilon/\omega = 2 \cdot 0.02/5 \times 10^{-4} = 80$ mm on the slides; 600 mm is not a multiple of 80, so we use $h = 75$ **mm, 9 stops per slide**. For the wrist the same bound between home and the end of the travel gives $5 \times 10^{-5} \times 2.36 = 1.2 \times 10^{-4}$ rad, well inside the budget, so the wrist distortion is modelled as the **homing offset plus a linear term** and needs no dense sampling. Our fitted model is a polynomial rather than an interpolating table, so the bound is not binding; we kept the spacing because it also resolves the distortion's slowest ripple (period ≥ 180 mm) and averages tracker noise along the axis. Section 6.5 shows that 150 mm would have worked as well.

The tracker's systematic field is the one error that *no* spacing improves: it is the same every time at the same place, up to 0.1 mm, and it limits what the calibration can achieve.

3.3 Readings per pose from the noise

We give tracker noise one tenth of the orientation budget, 10^{-4} rad, so $M = \lceil (5.8 \times 10^{-4}/10^{-4})^2 \rceil = \mathbf{34}$ readings per pose ($\sigma/\sqrt{M}$), taken as 5 commands × 7 readings = 35. A reading costs 0.03 s and a re-command 0.05 s, so a pose costs 1.3 s of sampling — far less than the 2–20 s moves. This sizing is done by `workflow.plan_calibration_poses` from `desired_accuracy`.

3.4 Error budget

Table 3: First-order error budget.

source (after calibration)	position	orientation
tracker systematic field (floor)	≤ 0.1 mm per point	≤ $\sim 10^{-4}$ rad (field gradient × body size)
joint random error (floor, at test time)	≈ 0.02 mm	≈ 2×10^{-4} rad
tracker noise, M = 34, averaged further over ~50 poses	≈ 0.01 mm	≈ 3×10^{-5} rad
model truncation (ripple not in the polynomial)	≤ 0.01 mm	≤ 10^{-4} rad

source (after calibration)	position	orientation
expected total	**≈ 0.1 mm typical, < 0.2 mm worst case**	**≈** 3×10^{-4} **rad, <** 10^{-3} **worst case**

4 Calibration method

4.1 Pose plan (method 1, step b)

The handout's method 1, with one change: the plan is *home, then sweep each joint on its own while the other joints rest at their home value*, and the robot does **not** return home between sweeps — the first stop of the next sweep moves the previous joint back in the same command (the move time is set by the slowest joint), which saves about 85 s of simulated time with no measurable loss of accuracy (we tested both). The plan for seed 1:

- one home pose;
- slides 1–3: 9 stops each, 75 mm apart (27 samples);
- wrist 4–6: stops at -2.00, -1.00, 1.00, 2.00 rad (12 samples);
- every pose is commanded 5 times in a row with 7 tracker readings each (35 readings per pose); all readings of a pose are averaged on the raw marker positions and registered once, so the fit sees one low-noise sample per pose;
- in total 40 poses / 200 commands, predicted 305 s (263 s motion + 42 s sampling), measured 305 s.

Joints 1–3 are kinematically decoupled from 4–6: a slide sweep moves the end effector along a straight line (direction = the real axis, distance = the real travel) with the orientation unchanged, and a wrist sweep rotates it about a fixed point (the wrist centre) about the real axis. Every parameter of the model is therefore excited by exactly one sweep, and the ten baseline poses of Step 6 (already paid for) are added to the data set.

Re-commanding a pose, as the handout suggests, costs only $t_{settle} = 0.05$ s and re-rolls the joint repeatability noise (10^{-4} rad per wrist joint — the same size as the tracker noise after averaging), so the 35 readings of a pose are split over 5 commands. Section 6.5 shows that this lowers the orientation error by about 30 % for 9 s of extra time.

4.2 Parameterization (method 1, step c)

`kinematics.ImprovedKinematics` (numerics in `src/student/_core.py`) uses the parameter vector

Table 4: Parameter vector (31 parameters; all zeros = the nominal robot at $\mathbf{F}_{X,init}$).

params	meaning	count
$\vec{\xi}_X$	twist of the base frame, $\mathbf{F}_X = \mathbf{F}_{X,init}\,\mathrm{Exp}(\vec{\xi}_X)$	6
(t_1, t_2) for $\vec{d}_2, \vec{d}_3, \vec{n}_4, \vec{n}_5, \vec{n}_6$	tilt in the plane perpendicular to the nominal axis	10
$\vec{\xi}_{we}$	twist of the tool frame, $\mathbf{F}_{we} = \mathbf{F}_{WE}\,\mathrm{Exp}(\vec{\xi}_{we})$	6
$c_{k,j}$	$\Delta\theta_k^{sys} = \mathrm{offset}_k + \sum_j c_{k,j}(\theta_k/\theta_k^{max})^j$, degree 2 (slides), 1 (wrist)	9

The joint **offsets are not fitted**: they come from `HomeRobot()`, which drives the robot to $\theta^* = 0$ and returns the encoders, $-\Delta\theta^{sys}(\mathrm{home}) - \Delta\theta^{rnd}$. Five homings cost $4\,t_{settle}$ after the first and

reduce the random part by $\sqrt{5}$. A slide offset is indistinguishable from a shift of the base along that axis, so it could not be fitted anyway. The wrist distortion is linear because the handout's slope bound (5×10^{-5}) limits any curvature to $\sim 10^{-4}$ rad over the travel; a quadratic wrist term only fits noise and made results worse (Section 6.5). The tilt of an axis has two parameters: a rotation about the axis itself does not change it.

4.3 Least-squares fit

For every sample j the residual is the pose error $\mathbf{D}_j = \left(\mathbf{F}_X \, \mathbf{F}_{ee}(\vec{\theta}_j)\right)^{-1} \mathbf{F}_{obs,j}$, stacked as $[\, w \,\mathrm{Log}(\mathbf{D}_j.R);\ \mathbf{D}_j.p \,]$ with $w = 200$ mm/rad so that the handout's 10^{-3} rad counts like its 0.2 mm. `fit` runs `scipy.optimize.least_squares` (Levenberg–Marquardt, central-difference Jacobian) from the nominal model with $\mathbf{F}_{X,init}$ from `estimate_base_frame` (the average of $\mathbf{F}_{obs,j}\,\mathrm{FwdKins}(\vec{\theta}_j)^{-1}$ over the samples). The real robot differs from the nominal one by only 1–2 %, so 5 function evaluations suffice; the final residuals are 0.029 mm rms / 0.077 mm max in position and 4.92e-04 rad rms in orientation — i.e. the model explains the measurements down to the tracker's own noise and field.

4.4 Identifiability (the Jacobian rank report)

`jacobian_rank_report` differentiates the residual numerically and reports the singular values of the Jacobian (columns scaled to a common unit). With **all 31 parameters free the rank is 30**: exactly one direction is invisible to the tracker, and its singular vector is `tilt_axis2_x`, `tool_alpha_x`, `tilt_axis3_x`, `base_alpha_x`, `tilt_axis6_x`, `tilt_axis5_x` with equal weights (Figure 2). This is the *gauge* of the base frame: rotating the base about $\vec{d}_1$ while tilting $\vec{d}_2, \vec{d}_3, \vec{n}_5, \vec{n}_6$ and the tool frame by the opposite angle about $\vec{d}_1$ produces identical tracker observations ($\vec{n}_4 \parallel \vec{d}_1$ is unaffected, and $\mathbf{F}_{cw}$ lies along $\vec{d}_1$). The tracker only ever sees the product $\mathbf{F}^*_{rob}\mathbf{F}^*_{ee}$; $\vec{d}_1$ fixes the base x axis and the wrist centre fixes the origin, but nothing fixes the rotation about x. We therefore **freeze** ξ_{X,α_x} (`fix_gauge=True`), which defines the base frame as "the x-rotation of $\mathbf{F}_{X,init}$"; the remaining 30 parameters have full rank with condition number 73. Predictions in the tracker frame, `predict_in_tracker`, do not depend on the gauge, and the handout's evaluation metric is formulated there.

Two more non-identifiable quantities are excluded by construction, as discussed above: the component of $\vec{\phi}_k$ along $\vec{d}_k$, and the slide offsets (versus the base position). Because of the gauge, *axes and tilts should be compared with ground truth through gauge-free quantities* — the angles between axes, the joint scale factors, and end-effector poses — or after removing the gauge rotation with the true base frame (`compare_with_ground_truth`), which is what Table 7 does.

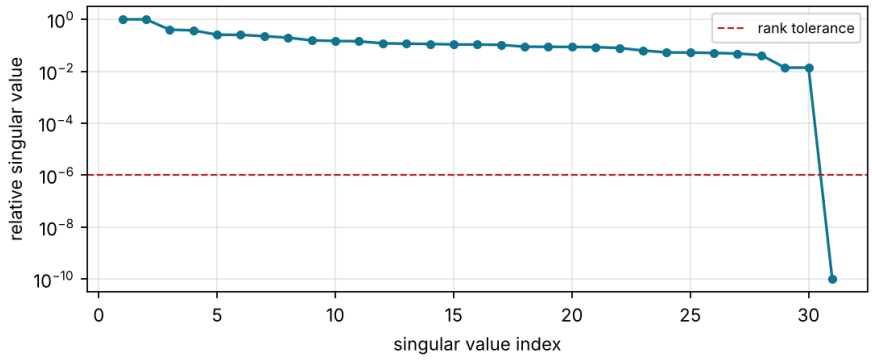


Figure 2: Singular values of the full Jacobian (all 31 parameters); the last one is the gauge direction.

4.5 Improved forward and inverse kinematics (method 1, step d)

`ImprovedFwdKins`: $\theta_k^* = \theta_k + \Delta\theta_k^{sys}(\theta_k)$, then the nominal chain with the calibrated axes and tool frame. `ImprovedInvKins` is the provided iterative solver (Levenberg–Marquardt on the pose error, both wrist branches, nearest in-limit solution); it works on our `fwd_kins` unchanged and converges in a few iterations because the model is within 1–2 % of the nominal one.

5 Software and unit tests

Table 5: Software layout.

module	contents
`src/student/design.py`	Step-1 design values (tracker pose, robot pose, marker layout, envelope)
`src/student/metrics.py`	`frame_error` (log map, exact for large angles), `summarize_errors` (+ p95, target fraction)
`src/student/registration.py`	Arun registration (own implementation; rejects < 3 or collinear points, fixes reflections), `average_frames` (chordal mean), `refine_marker_body`
`src/student/workflow.py`	`measure_home_offsets`, `plan_calibration_poses` (sizing from accuracy, visibility with margin, time prediction), `collect_calibration_data`, `estimate_base_frame`, `merge_data`, `order_poses_by_motion_time`
`src/student/kinematics.py`	`ImprovedKinematics`: parameterization, `fwd_kins`, `residuals`, `fit`, `jacobian_rank_report`, `parameters_report`, `compare_with_ground_truth`
`src/student/_core.py`	pure-numpy se(3) math, the parameter model, Jacobian, least squares, rank analysis, plan-sizing formulas
`src/student/_convert.py`	conversions between the course's `Frame`/`Rot`/`vct3` and numpy arrays
`report/variant_study.py`	the 10-seed pose-plan trade-off study of Section 6.5
`tests/test_student.py`	30 unit tests (below)

`tests/test_student.py` keeps every test the course shipped, implements the three it left as `TODO`, and adds tests for the remaining components. All 30 tests pass (53 passed in 18 s (23 provided + 30 ours), run on 2026-10-31).

Table 6: Unit tests and their pass criteria (grouped; 30 test functions).

#	component	test	pass criterion
1	`frame_error`	known values; invariance to a common left factor; 2.5 rad rotation; offset measured in the reference frame	exact to 10^{-9}
2	`summarize_errors`	rms / max / target fraction on hand-made lists; empty input	exact
3	`register_points`	recovers a random frame from 6 points	10^{-9}
4	`register_points`	2 points, collinear points, mismatched lengths	`ValueError`
5	`register_points`	300 noisy registrations of the 8-marker body	rms position 0.03–0.09 mm, orientation $(0.5\text{–}1.5) \times 10^{-3}$ rad, proper rotation
6	`average_frames`	identical frames; proper rotation; 100 noisy frames	exact; $R^T R = I$; noise reduced to ~1/10
7	`refine_marker_body` (TODO)	body with manufacturing error, 30 readings	layout error shrinks > 3×, to < 0.15 mm; costs 30 × 0.03 s
8	design / visibility	envelope corners visible, also shifted by ±85 mm on every axis	all visible; 85 mm > worst-case base shift
9	`plan_calibration_poses` (TODO)	every planned pose within limits and visible with margin; every joint swept; 34 → 5 × 7 readings, h = 75 mm, 40 poses / 200 commands; impossible margin	`ValueError`
10	plan sizing	tighter accuracy → more readings and more time	monotone
11	`measure_home_offsets`; `order_poses_by_motion_time`	20 homings vs the next homing's encoders; cost of 5 homings; vs true $\Delta\theta^{sys}(0)$; ordering 12 random poses shortens the travel	within $4\sigma_{rnd}$; one move + 4 t_{settle}
12	`collect_calibration_data`	commands A, B, B × 4 readings: B merged into one sample with 8 readings, encoder values, clock cost (3 moves + 12 × 0.03 s), frames near truth	exact clock; < 0.3 mm / 2×10^{-3} rad
13	`collect_calibration_data`	pose outside the frustum	skipped, recorded
14	`estimate_base_frame`	perfect nominal data; perfect data from a perturbed improved model after `merge_data`	exact
15	`ImprovedKinematics`	zero parameters = nominal kinematics; `inv_kins` round trip with perturbed parameters	10^{-9}; 10^{-6} mm / 10^{-8} rad
16	parameter layout	offsets at home; polynomial value and scale; a 0.01 rad tilt of $\vec{d}_2$ moves a 300 mm travel by 3 mm	exact
17	`residuals`	zero on self-generated data; 10^{-3} rad shows up as 0.2 mm-equivalent	10^{-9}

#	component	test	pass criterion
18	`fit` (TODO)	noise-free data from a known non-zero parameter vector (gauge included), fitted from zero	poses reproduced to 10^{-5} mm; parameters differ (gauge), axis angles and scales agree
19	`jacobian_rank_report`	rank $31-1$ with all parameters; weakest direction = gauge; full rank after fixing it	condition < 10^4
20	end to end	full pipeline on seed 0; 150 random poses vs `actual_ee_pose`; parameters vs truth (gauge-free)	< 0.2 mm, < 10^{-3} rad; t_{cal} < 900 s; axis errors < 0.5 mrad; distortion functions within the tracker floor

6 Results

All numbers below come from `notebooks/assignment2_solution.ipynb` (`results/notebook_results.json`). Ground truth (`rob.actual_ee_pose`, `rob.ground_truth_parameters`) is used only in the evaluation and reporting steps, never in the calibration.

6.1 Calibrated parameters versus ground truth (seed 1)

Table 7: Parameter errors; the true angles are 90.0685, 90.1799, 88.3080, 91.8122, 90.1719, 89.3844 degrees.

quantity	estimated − true
angles between axes (d1–d2, d1–d3, d2–d3, n4–n5, n4–n6, n5–n6), mrad	0.056, 0.092, -0.027, -0.038, 0.066, -0.072
axis direction errors after removing the base gauge (joints 1–6), mrad	0.040, 0.069, 0.067, 0.080, 0.086, 0.094
joint scale factors vs the best straight line through the true distortion (1–3 mm/mm; 4–6 rad/rad)	6.7e-05, -2.6e-05, -5.0e-05, 2.3e-05, 5.5e-05, 7.3e-05
joint distortion functions over the travel, max deviation (1–3 mm; 4–6 rad)	2.9e-02, 2.9e-02, 2.3e-02, 7.6e-05, 1.5e-04, 2.5e-04
joint offsets at home (joints 1–3, mm; 4–6, rad)	-4.7e-03, 7.2e-04, 1.3e-03, -2.2e-05, 2.3e-05, -2.3e-05
tool frame (rad, mm)	1.06e-04, 0.014
base frame in the tracker (rad, mm; includes the gauge)	5.82e-03, 0.090

The axes are recovered to a tenth of a milliradian, the wrist distortion to $\sim 10^{-4}$ rad and the offsets to the averaged repeatability noise. The slide distortion functions agree with the truth to a few hundredths of a millimetre: the fit cannot distinguish a slide's scale error from the tracker field's variation along that slide, so this residual is the field's (≤ 0.1 mm) — it is consistently reproduced in the tracker frame and does not hurt the pose accuracy. (The handout's "joint scale" s_k is not directly comparable with a fitted linear coefficient because the ripple $\sigma \sin(A\theta + B)$ has a local slope of up to $\sigma A = 3.5 \times 10^{-4}$; we therefore compare with the slope of the best straight line through the true distortion.) The base frame error is larger than the pose errors because the base origin lies 500 mm outside the measured region (it is extrapolated from the wrist centre along $\vec{d}_1$) and because it contains the gauge rotation.

6.2 Forward kinematics over the work volume (Step 9)

Fixed test grid: $3^6 = 729$ joint vectors (slides at 100/300/500 mm and −200/0/200 mm, wrist at ±1.3 rad and θ_6 at ±2.0 rad, all visible). Metric: $\mathbf{D} = (\mathbf{F}_{trk}^{-1}\mathbf{F}_{ee}^{*})^{-1}(\widehat{\mathbf{F}}_X\,\text{ImprovedFwdKins})$.

Table 8: `ImprovedFwdKins` versus the nominal kinematics on the test grid (targets 0.2 mm, 10^{-3} rad).

seed	model	pos rms [mm]	pos max [mm]	ang rms [mrad]	ang max [mrad]	within target	target met
1	nominal	7.070	16.104	32.932	55.990	0%	no
1	**improved**	**0.080**	**0.118**	**0.211**	**0.348**	100%	**yes**
2	nominal	5.562	12.611	35.526	53.614	0%	no
2	**improved**	**0.062**	**0.108**	**0.235**	**0.445**	100%	**yes**
3	nominal	6.122	13.191	21.582	32.527	0%	no
3	**improved**	**0.065**	**0.109**	**0.347**	**0.572**	100%	**yes**

6.3 Inverse kinematics (Step 10)

40 reachable targets (true poses of random test-grid joint vectors). For each, $\vec{\theta} = \text{ImprovedInvKins}(\widehat{\mathbf{F}}_X^{-1}\mathbf{F})$, the robot is moved there and the actual placement of the calibration body is compared with the target. Every real move includes the joint repeatability noise ($\sigma = 0.005$ mm, 10^{-4} rad), which puts a floor of ≈ 0.02 mm / 2×10^{-4} rad under this number.

Table 9: `ImprovedInvKins` versus the nominal inverse kinematics (40 targets per seed).

seed	model	pos rms [mm]	pos max [mm]	ang rms [mrad]	ang max [mrad]	within target	target met
1	nominal	7.432	13.274	34.505	55.931	0%	no
1	**improved**	**0.083**	**0.115**	**0.292**	**0.494**	100%	**yes**
2	nominal	5.957	11.596	35.738	52.121	0%	no
2	**improved**	**0.066**	**0.098**	**0.291**	**0.609**	100%	**yes**
3	nominal	6.367	12.402	21.589	32.167	0%	no
3	**improved**	**0.066**	**0.088**	**0.372**	**0.742**	100%	**yes**

6.4 Time

Table 10: Simulated time (soft budget 900 s). The baseline poses are the template's Step 6. The IK test's 80 moves (40 targets × nominal and improved) are visited in a greedy nearest-neighbour order, which halves the testing time of the template's random order (≈ 1080 s) — the wrist moves at 0.1 rad/s.

seed	homing + baseline [s]	sweeps [s]	**calibration total [s]**	testing (80 moves) [s]	commands	readings
1	84	305	**389**	481	200	1400
2	84	305	**389**	475	200	1400
3	84	305	**389**	458	200	1400

The whole notebook, including the three-seed repetition, runs in well under the 10-minute real-time limit (see the notebook's last printed line); the 10-seed variant study is a separate script (about 6 minutes).

6.5 Pose-plan / time trade-off

Same pipeline with deliberately different plans, each on 10 robot seeds (`report/variant_study.py`; forward error on the test grid, no baseline poses; rms values are means over the seeds with their standard error, max values are the worst seed):

Table 11: Design variants (10 seeds each; every variant met both targets on every seed).

plan	t_{cal} [s]	cmds	rdgs	pos rms [mm]	pos max [mm]	ang rms [mrad]	ang max [mrad]
default: wrist ±1,±2; h 75; 5×7 readings	290	200	35	0.062	0.108	0.120 ± 0.01	0.305
1 command × 34 readings	281	40	34	0.063	0.117	0.174 ± 0.02	0.450
wrist ±0.5,±1 rad	202	200	35	0.065	0.124	0.211 ± 0.02	0.527
wrist ±0.75,±1.5 rad	245	200	35	0.065	0.118	0.154 ± 0.01	0.328
wrist ±1.15,±2.3 rad	317	200	35	0.063	0.106	0.123 ± 0.01	0.251
slide step 150 mm	274	140	35	0.065	0.106	0.124 ± 0.01	0.269
5×2 = 10 readings per pose	260	200	10	0.066	0.144	0.225 ± 0.02	0.504
5×20 = 100 readings per pose	368	200	100	0.064	0.098	0.105 ± 0.01	0.267
wrist polynomial degree 2	290	200	35	0.062	0.107	0.140 ± 0.01	0.280
slide polynomial degree 1	290	200	35	0.064	0.110	0.143 ± 0.01	0.316

What the table shows. (i) The **position** error is the same for every plan, ≈ 0.063 mm rms: it is set by the tracker's systematic field, not by the plan. (ii) The **wrist sweep length** is the main lever on orientation: ±1 rad is almost twice as bad as ±2 rad, ±1.5 rad is in between, and ±2.3 rad brings nothing more — the axis-direction error falls as $1/\Theta$ until the tracker's own orientation floor is reached. (iii) **Re-commanding** each pose 5× (same number of readings) lowers the orientation rms by ~30 % for 9 s: it averages the joint repeatability noise, which a single command cannot. (iv) **Readings per pose** follow $1/\sqrt{M}$: 10 readings are clearly worse, 100 only slightly better than 35 at +78 s. (v) The **slide spacing** does not matter for a polynomial model (150 mm is as good as 75 mm and 16 s faster; we kept 75 mm for the reasons in Section 3.2). (vi) A **quadratic wrist** term and a **linear-only slide** term are both worse than our (2, 2, 2, 1, 1, 1) choice, confirming the error model. The default plan sits at the knee of the time/accuracy curve.

7 Discussion

What limits the accuracy. After calibration the residual error is dominated by the two floors that no model can remove: the tracker's systematic field (≤ 0.1 mm, slowly varying, so it leaks a fraction of itself into every parameter and into every "bottom line" tracker test) and, in the inverse-kinematics test, the joint repeatability noise re-rolled at every move. The rms forward error of ≈ 0.080 mm is consistent with the budget of Table 3.

Identifiability shaped the design. The rank report confirmed the gauge direction and led us to (a) fix one base-twist component, (b) take the joint offsets from homing instead of the fit, and (c) parameterize tilts with two components. Comparing parameters with ground truth has to respect the gauge: raw tilt vectors disagree by a rotation about $\vec{d}_1$ even when the model is perfect, which is why Table 7 reports gauge-free quantities.

Wrist singularity. At $\theta_5 = \pm\pi/2$ (inside the ±0.75π range) axes 4 and 6 align and the inverse kinematics is ill-conditioned; the provided solver handles it with damping and by trying both wrist branches. The test grid avoids the exact singularity (±1.3 rad).

Deviations from the template. `plan_calibration_poses` accepts `return_info=True` and returns its sizing (readings per command, predicted time); the notebook passes that number as `repeats` to `collect_calibration_data` instead of the template's `repeats=1`, and `collect_calibration_data` averages the readings of consecutive identical commands into one sample; the Step-6 baseline data are merged into the fit (`workflow.merge_data`); the IK test visits its 40 targets in a travel-minimising order (`workflow.order_poses_by_motion_time`); design values live in `src/student/design.py`; the robot does not return home between sweeps. Everything else keeps the template's signatures.

What we would do next. A denser bottom-line tracker test ordered to minimise wrist travel; a per-seed check of the variance of the fitted parameters from the Jacobian ($\sigma^2(J^TJ)^{-1}$) as a confidence measure; and, if the tracker field were calibratable (it is not, with the given API), a low-order spatial correction of it.

8 Conclusion

Method 1 — home, sweep each joint alone, fit a 31-parameter model by least squares with the joint offsets taken from homing — meets the handout's targets with margin on every seed tried (0.118 mm / 3.48e-04 rad forward, 0.115 mm / 4.94e-04 rad inverse on seed 1) in about 389 s of simulated time, less than half of the soft budget, and the program runs in well under the 10-minute limit.

9 Appendix: how to run

```
cd assignment2
python -m pytest tests -q            # all unit tests
jupyter notebook notebooks/assignment2_solution.ipynb
#   run top to bottom; writes results/notebook_results.json
python report/make_report.py         # regenerates this report (needs pandoc)
```